

INDIVIDUAL SEMESTER PROJECT · MILESTONE 1 TECHNICAL REPORT

PREPARED FOR
Molemane Community Newspaper
Lichtenburg · Media Industry

Student	Sentse, X
Student Number	37940066
Project ID	CMPG325-2026-128
Client ID	CLI-128
Assigned Addressing Block	172.30.84.0/23
Assigned Networking Challenge	SSH (secure device management plane)
Milestone 1 Due	28 August 2026
GITHUB	Xoli020525

1. Executive Summary

Molemane Community Newspaper is a small media company located in Lichtenburg, with departments responsible for administration, editorial work, advertising, and design/production. The organisation also has a boardroom used for client and management meetings. This report outlines the Milestone 1 requirements for the CMPG 325 individual semester project, including an assessment of the client's needs, a proposed physical network topology, a logical topology consisting of eight VLANs, and a VLSM IP addressing plan based on the allocated 172.30.84.0/23 address block.

The network design focuses on three main requirements identified in the project brief: providing suitable connectivity and network services for the different departments, providing the boardroom with both dedicated wired presentation access and wireless connectivity, and preparing for two technical requirements that will be implemented later. These include the SSH device-management requirement in Milestone 2 and the CR11 requirement for VoIP handsets. The necessary VLANs and IP address space for these requirements have therefore been planned in advance to avoid major changes to the network design later.

2. Client Requirements Analysis

Requirements were extracted directly from the project brief (CMPG325-2026-128) and translated into concrete design decisions, summarized below.

#	Requirement	Source	Design Response
1	Connectivity and appropriate network services for all newspaper departments	Client brief	Hierarchical topology: edge router, L3 core switch, dedicated access switch per department
2	Use assigned block 172.30.84.0/23 as the addressing basis	Client brief	VLSM subnetting into 8 VLANs + WAN link, ~60% of the /23 reserved for growth
3	Boardroom needs dedicated wireless AND wired presentation ports	Design constraint	VLAN 50 with one wired presentation drop and one dedicated AP/SSID on SW-BOARD
4	Configure, verify and demonstrate SSH device management	Networking challenge	VLAN 99 management plane; SSH-only VTY lines on R1 and all switches
5	Accommodate VoIP handsets without redesigning the network (CR11)	Change request	Dedicated VLAN 70 for voice, isolated from data traffic, sized for growth
6	Produce a working, testable Packet Tracer implementation	Client brief	Addressed in Milestone 2 build phase, on top of this Milestone 1 design

Table 1 — Client requirements traced to design responses.

Molemane Community Newspaper's operational needs map naturally onto five functional groups plus supporting infrastructure:

- Administration & Management: office administration, finance, HR-type functions; moderate traffic, needs access to shared file/print services.
- Editorial / Newsroom: the largest user group; journalists and editors filing stories, needing reliable access to file storage and for research.
- Advertising & Sales: client-facing staff selling ad space; similar profile to admin, moderate headcount.

- Design & Pre-Press Production: page layout and print preparation; smaller but performance-sensitive group.
- Boardroom: shared meeting space used by management and, according to the design constraint, requiring simultaneous dedicated wired and wireless presentation connectivity.
- Supporting infrastructure: a small server function (file/print/DNS), a device-management plane for secure SSH access, and a voice VLAN reserved for the CR11 VoIP roll-out.

3. Network Design Overview

A hierarchical core–access network design was selected instead of a flat, single-switch design because it separates the broadcast domains of each department, allows the network to expand easily if Molemane hires more staff or opens another office, and provides a central point at SW-CORE for inter-VLAN routing and future ACL/security policies. An L3 multilayer switch handles inter-VLAN routing through SVIs rather than using a router-on-a-stick approach. This makes the network more efficient as the number of VLANs increases and prevents a single router link from becoming a bottleneck.

Each department has its own access-layer switch, which connects to SW-CORE using an 802.1Q trunk. This design helps prevent problems on one department's switch from affecting the traffic of other departments and also reflects the organization's physical layout, such as separate offices or floors. The boardroom has its own access switch so that its dedicated wired presentation connection and wireless access can be provided without placing boardroom traffic on another department's VLAN.

4. Physical Topology

Figure 1 shows the proposed physical arrangement of devices: an internal boundary router (R1), an L3 core switch, five department/purpose access switches, a small server function attached to the core, and the boardroom's dedicated wired presentation port and access point.

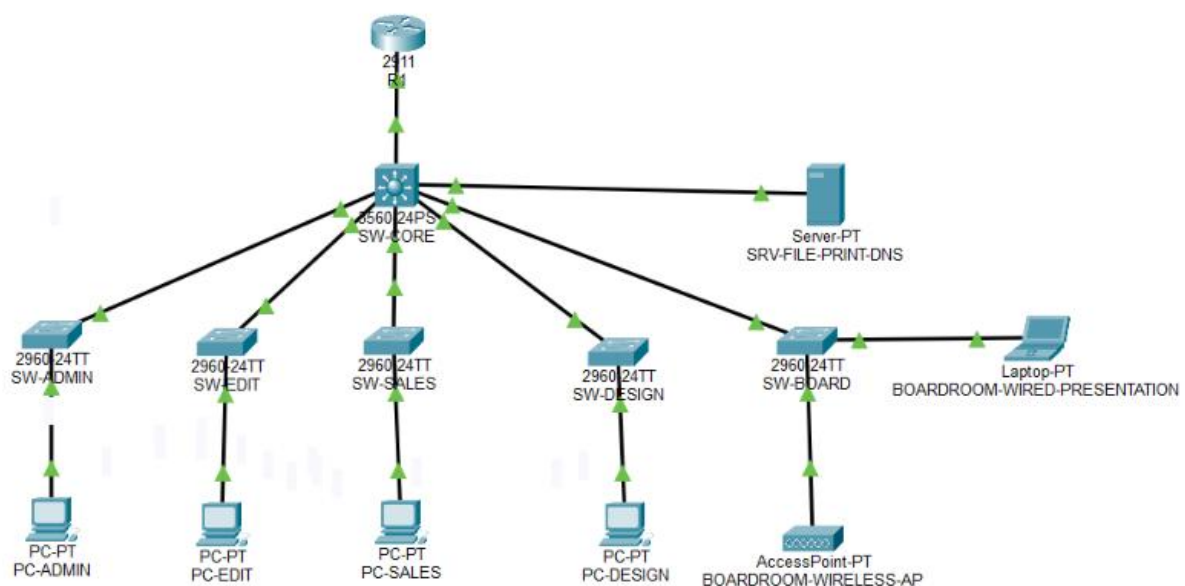


Figure 1 — Proposed physical topology for Molemane Community Newspaper.

Design notes:

- R1 functions as the network's internal boundary router, sitting between SW-CORE and a reserved uplink port. The network operates as a closed, internally-routed system with no Internet connection; R1's uplink interface is provisioned in the addressing plan but is not connected to any external network.
- SW-CORE performs all inter-VLAN routing and is the only device with SVIs; access switches carry VLAN traffic on trunk uplinks only.
- The server function is attached directly to SW-CORE to minimize latency to file/print/DNS services used by every department.
- SW-BOARD carries two dedicated links out to the boardroom itself: one wired presentation drops and one link to a dedicated access point, satisfying the stated design constraint without sharing the link with any other department.

5. Logical Topology and VLAN Design

Figure 2 illustrates the logical structure of the network, showing the eight VLANs, their subnets, SVI gateway addresses, and the address space reserved for future expansion. VLAN 99 is dedicated to network device management and will be used for the SSH traffic required by the assigned networking challenge. It is intended only for network devices and does not include end-user hosts.

VLAN 70 has been included in the design in preparation for the future VoIP deployment. This allows the CR11 requirement to be implemented later by adding the VoIP handsets and configuring the appropriate voice VLAN on the access ports, without requiring changes to the existing IP addressing scheme.

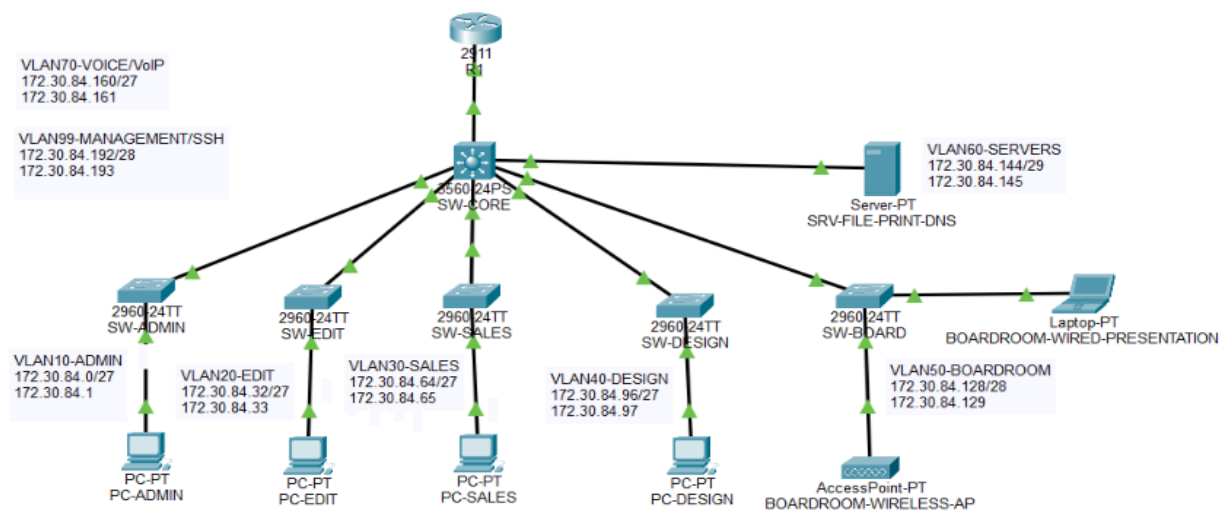


Figure 2 — Logical topology: VLANs, subnets, and SVI gateways

VLAN	Name	Access Switch	Port Type
10	ADMIN	SW-ADMIN	Access (data)
20	EDITORIAL	SW-EDIT	Access (data)
30	SALES	SW-SALES	Access (data)
40	DESIGN	SW-DESIGN	Access (data)
50	BOARDROOM	SW-BOARD	1x access (wired presentation) + 1x access to AP (wireless)
60	SERVERS	SW-CORE (directly attached)	Access (data)
70	VOICE	All access switches (auxiliary VLAN)	Voice VLAN on phone-capable ports
99	MANAGEMENT	All switches + R1	SVI / management interface only, no end-user ports

Table 2 — VLAN allocation summary.

6. IP Addressing Plan (VLMS)

The allocated address block 172.30.84.0/23 contains 512 IP addresses, ranging from 172.30.84.0 to 172.30.85.255. VLSM was applied to allocate appropriately sized subnets to each VLAN according to its expected number of hosts, rather than assigning the same subnet size to every VLAN. This approach makes better use of the available address space while keeping most of the block available for future expansion, such as additional departments, another office, or new VLANs.

VLAN	Department / Purpose	Subnet	Mask	Gateway (SVI)	Usable Range	Hosts Req'd
10	Administration & Management	172.30.84.0/27	255.255.255.224	172.30.84.1	.2 – .30	12
20	Editorial / Newsroom	172.30.84.32/27	255.255.255.224	172.30.84.33	.34 – .62	25
30	Advertising & Sales	172.30.84.64/27	255.255.255.224	172.30.84.65	.66 – .94	18
40	Design & Pre-Press Production	172.30.84.96/27	255.255.255.224	172.30.84.97	.98 – .126	12
50	Boardroom (wired + wireless)	172.30.84.128/28	255.255.255.240	172.30.84.129	.130 – .142	10
60	Servers (File / Print / DNS)	172.30.84.144/29	255.255.255.248	172.30.84.145	.146 – .150	5
70	Voice / VoIP handsets (CR11)	172.30.84.160/27	255.255.255.224	172.30.84.161	.162 – .190	25 (reserve)
99	Device Management Plane (SSH)	172.30.84.192/28	255.255.255.240	172.30.84.193	.194 – .206	13 (devices)
—	R1 boundary link (reserved, not connected)	172.30.84.208/30	255.255.255.252	172.30.84.209	.209 – .210	2
—	Reserved for future growth	172.30.84.224/27 — 172.30.85.255/24	—	—	unused block	—

Table 3 — VLSM addressing plan derived from 172.30.84.0/23.

Sizing Rationale: The Editorial department was assigned the largest data subnet, a /27 with 30 usable addresses, because it has the highest number of users. The servers network was allocated a /29 subnet, as only a small number of infrastructure devices are expected to use it. VLAN 99 (Management/SSH) and VLAN 50 (Boardroom) were each assigned /28 subnets, providing sufficient addresses for management interfaces, access points, and boardroom devices without unnecessary address wastage. R1's boundary link uses a /30 subnet, which is the smallest subnet suitable for a point-to-point connection. However, this link remains unconnected because the current network does not have an external uplink.

7. Accommodating the Design Constraint and Change Request

7.1 Design constraint — dedicated boardroom connectivity

VLAN 50 (172.30.84.128/28) is reserved specifically for the boardroom. SW-BOARD provides one access port connected directly to a boardroom wall point for laptop or presentation use, as well as a separate access port connected to a dedicated access point for the boardroom's wireless SSID. Both connections use the same VLAN and subnet, allowing presenters to receive the same network access whether they connect through the wired or wireless option. The boardroom connections are also kept separate from the switches used by other departments.

7.2 CR11 — VoIP handset accommodation

To avoid making changes to the IP addressing plan later, VLAN 70 (172.30.84.160/27) has been reserved for voice traffic. The subnet can support up to 25 VoIP handsets, while also allowing room for future growth. When the handsets are introduced, the relevant access ports will use VLAN 70 as the auxiliary voice VLAN alongside each department's existing data VLAN. This keeps voice and data traffic logically separated while allowing both to use the same physical network connection, without requiring additional cabling or IP re-addressing.

7.3 Assigned challenge — SSH device management (preview)

VLAN 99 (172.30.84.192/28) is reserved as the management network for R1 and all switches. In Milestone 2, the network devices will be configured for secure SSH access by setting a domain name, generating 2048-bit RSA keys, creating a local administrator account, and restricting the VTY lines to SSH using transport input ssh. Telnet will also be disabled on all devices. The configuration will then be verified by connecting to the devices through SSH from a management host on VLAN 99 and confirming that Telnet connections are rejected.